Design and Analysis of Algorithms (E1ITB303)

Micro Project Report

On

AI-Based Smart Allocation Engine for PM Internship Scheme

Submitted by

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CERTIFICATE

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Abstract

The allocation of students to internship opportunities is a complex and critical problem in large-scale academic and professional ecosystems. With the increasing number of applicants and limited internship positions, manual or rule-based allocation systems often fail to ensure fairness, efficiency, and optimal utilization of available resources. This problem becomes even more challenging when multiple constraints such as student preferences, eligibility criteria, and internship capacity limits must be simultaneously satisfied.

This project presents the design and implementation of an AI-based Smart Allocation Engine that models the internship assignment process as a constrained optimization problem. The system leverages core concepts from Design and Analysis of Algorithms (DAA), including Greedy algorithms, Backtracking, and the Assignment Problem framework, to generate efficient and optimal allocation solutions.

Two distinct algorithmic approaches are implemented and analyzed. The first approach uses a Greedy strategy, which prioritizes students based on predefined criteria such as academic performance or ranking and assigns internships by selecting the best available option at each step. While this approach is computationally efficient and scalable, it may not always produce the globally optimal solution. The second approach employs a Backtracking technique, which systematically explores all possible allocations to identify the optimal solution that maximizes overall satisfaction. Although computationally expensive, this method guarantees optimality.

To evaluate the effectiveness of the allocation strategies, a preference-based scoring system is introduced, where higher preference matches yield higher scores. Additionally, performance metrics such as execution time, allocation efficiency, and fairness are analyzed to compare the two approaches. The results demonstrate a clear trade-off between computational efficiency and optimality, highlighting the importance of selecting appropriate algorithms based on application requirements.

This project not only demonstrates the practical application of algorithmic techniques to a real-world problem but also provides a scalable foundation for developing intelligent allocation systems. Such systems can be extended to national-level platforms, including internship schemes and job allocation portals, where fairness, efficiency, and scalability are of paramount importance.

1. INTRODUCTION

1.1 Project Overview

In modern academic and professional environments, internship programs play a vital role in bridging the gap between theoretical knowledge and practical experience. However, allocating students to internships in a fair and efficient manner remains a challenging problem, especially when the number of applicants significantly exceeds the available opportunities. Traditional allocation methods, which often rely on manual decision-making or simple rule-based systems, are inefficient, error-prone, and incapable of handling large-scale data.

The internship allocation problem can be viewed as a combinatorial optimization problem, where multiple students compete for a limited number of internship positions, each with its own constraints and requirements. The complexity of the problem increases due to factors such as diverse student preferences, varying eligibility criteria, and limited capacity for each internship.

This project aims to address these challenges by developing an algorithm-driven smart allocation system that automates the assignment process. By leveraging advanced algorithmic techniques, the system ensures that constraints are satisfied while maximizing overall satisfaction among students. The project integrates theoretical concepts from Design and Analysis of Algorithms with practical implementation to create a solution that is both efficient and scalable.

1.2 Objectives of the Project

The primary objective of this project is to design and implement an intelligent system capable of allocating students to internship positions in an optimal and efficient manner. The key objectives include:

- To model the internship allocation problem as a formal optimization problem
- To implement Greedy and Backtracking algorithms for solving the problem
- To ensure that all constraints, including student preferences, eligibility criteria, and internship capacities, are satisfied
- To develop a scoring mechanism that quantifies the quality of allocation

- To compare the performance of different algorithmic approaches based on execution time and solution optimality
- To demonstrate the applicability of algorithmic techniques in solving real-world problems

1.3 Problem Statement

The problem involves assigning a set of students to a set of available internship positions based on multiple constraints. Each student provides a list of preferred internships ranked in order of priority, and each internship has a limited number of available slots. Additionally, students must satisfy specific eligibility criteria to be considered for certain internships.

The objective is to determine an assignment that maximizes the overall satisfaction score while ensuring that:

- Each student is assigned to at most one internship
- The number of students assigned to an internship does not exceed its capacity
- All eligibility constraints are strictly enforced

This problem is closely related to the classical assignment problem and can be extended to more complex real-world scenarios involving multiple constraints and optimization objectives.

1.4 Scope of the Project

The scope of this project includes the design, implementation, and analysis of algorithmic solutions for the internship allocation problem in a simulated environment. The system is capable of handling small to medium-sized datasets and provides a comparative analysis of different algorithmic approaches.

In a broader context, the proposed system can be extended to support large-scale applications such as university placement systems, government internship schemes, and job allocation platforms. Future enhancements may include the integration of machine learning techniques for predicting student preferences, real-time data processing, and deployment as a web-based application accessible to a large number of users.

1.5 Scope Expansion

Current:

Simulated environment

Algorithm comparison

Future:

Integration with government portals

Real-time allocation engine

AI-based recommendation system

2. ALGORITHM DESIGN

2.1 Problem Classification

The internship allocation problem can be classified as a combinatorial optimization problem, where the objective is to find the best possible assignment among a large set of feasible solutions. It is closely related to the classical assignment problem, which involves matching elements from two sets in a way that maximizes or minimizes a given objective function.

As the number of students and internships increases, the number of possible assignments grows exponentially, making the problem computationally intensive. In such cases, the problem may exhibit characteristics of NP-Hard problems, where finding the optimal solution within a reasonable time becomes challenging.

2.2 Greedy Algorithm

The Greedy algorithm is a heuristic-based approach that constructs a solution by making a sequence of locally optimal choices at each step. In the context of internship allocation, the Greedy algorithm works by first sorting the students based on their priority, which may be determined by factors such as academic performance or ranking. The algorithm then iterates through each student

and assigns them to the highest-ranked internship in their preference list that still has available capacity.

This approach is efficient because it reduces the computational complexity by avoiding the exploration of all possible combinations. However, the main limitation of the Greedy algorithm is that it does not consider the global impact of its decisions. As a result, it may produce suboptimal solutions in certain cases where a locally optimal choice leads to a globally inferior outcome.

2.3 Backtracking Algorithm

The Backtracking algorithm is an exhaustive search technique that systematically explores all possible solutions to find the optimal one. It can be visualized as a state-space tree, where each level represents a student and each branch represents a possible assignment of that student to an internship.

The algorithm starts with an empty assignment and recursively assigns internships to students while ensuring that all constraints are satisfied. If at any point a constraint is violated, the algorithm backtracks by undoing the last assignment and trying a different option. This process continues until all possible assignments have been explored.

Although Backtracking guarantees the optimal solution, it has a high computational complexity, especially for large datasets, as it explores all possible combinations. Therefore, it is more suitable for smaller problem sizes or scenarios where optimality is critical.

3. COMPLEXITY ANALYSIS

3.1 Theoretical Time Complexity Analysis

The efficiency of an algorithm is primarily determined by its time complexity, which represents how the execution time grows with the size of the input. In the context of this project, two different algorithmic approaches are analyzed: the Greedy algorithm and the Backtracking algorithm, each exhibiting significantly different time complexities.

The Greedy algorithm involves two main steps: sorting the students based on priority and assigning internships based on preferences. The sorting step typically requires O(nlog⁡n)O(n \log

n)O(nlogn) time, where nnn is the number of students. Following this, the algorithm iterates through each student and checks their list of preferred internships. In the worst case, each student may need to check all available internships, resulting in a time complexity of O(n×m)O(n \times m)O(n×m), where mmm is the number of internships. Therefore, the overall time complexity of the Greedy approach can be expressed as O(nlog⁡n+n×m)O(n \log n + n \times m)O(nlogn+n×m), which is considered polynomial and efficient for large datasets.

In contrast, the Backtracking algorithm explores all possible assignments of students to internships. Since each student can potentially be assigned to multiple internships, the total number of possible combinations grows exponentially with the number of students. In the worst-case scenario, the algorithm may need to evaluate n!n!n! (factorial) combinations, leading to a time complexity of O(n!)O(n!)O(n!). This makes Backtracking computationally expensive and impractical for large-scale problems, although it guarantees an optimal solution.

Algorithm	**Steps Involved**	**Practical Time Behavior**
Greedy	Sorting + Single pass allocation	Fast, increases slowly with input size
Backtracking	Recursive exploration of all combinations	Very slow, increases rapidly

---

3.2 Space Complexity Analysis

In addition to time complexity, space complexity is another important factor that determines the efficiency of an algorithm. The Greedy algorithm requires relatively minimal additional memory, as it primarily uses arrays or lists to store student preferences and internship capacities. Therefore, its space complexity is approximately O(n+m)O(n + m)O(n+m), which is manageable even for larger inputs.

The Backtracking algorithm, on the other hand, requires additional memory to store recursive function calls and temporary assignments. Since it explores all possible combinations, it must maintain a stack of recursive calls, which can grow up to O(n)O(n)O(n) in depth. Additionally, it

may store intermediate solutions, leading to increased memory usage. As a result, the space complexity of the Backtracking approach is higher compared to the Greedy algorithm.

Algorithm	Memory Usage	Reason
Greedy	Low	Stores only final allocation and capacity tracking
Backtracking	Medium	Stores recursive calls and multiple intermediate assignments

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3.3 Scalability Analysis

Scalability refers to the ability of an algorithm to handle increasing input sizes efficiently. The Greedy algorithm is highly scalable due to its polynomial time complexity, making it suitable for real-world applications involving thousands of students and internships. It can produce reasonably good results within a short time, making it ideal for real-time systems.

In contrast, the Backtracking algorithm does not scale well due to its exponential time complexity. As the number of students increases, the computation time grows rapidly, making it unsuitable for large-scale applications. However, it remains useful for smaller datasets or scenarios where achieving the optimal solution is critical.

Factor	Greedy Algorithm	Backtracking Algorithm
Handles large data	Yes	No
Suitable for real-time	Yes	No
Performance with increase in input	Stable	Degrades rapidly

3.4 Trade-off Between Efficiency and Optimality

One of the key insights from this analysis is the trade-off between efficiency and optimality. The Greedy algorithm prioritizes speed and scalability but may produce suboptimal solutions in certain cases. On the other hand, the Backtracking algorithm guarantees optimal results but at the cost of significantly higher computation time. This trade-off highlights the importance of selecting the appropriate algorithm based on the specific requirements of the problem.

4. IMPLEMENTATION

4.1 COMPLETE JAVA CODE

```
import java.util.*;

class Student {
    int id;
    int priority;
    List<Integer> preferences;

    Student(int id, int priority, List<Integer> preferences) {
        this.id = id;
        this.priority = priority;
        this.preferences = preferences;
    }
}

public class InternshipAllocation {

    static int bestScore = 0;
```

```
    static int[] bestAssignment;

    // -------- GREEDY ALGORITHM --------
    public static int[] greedy(List<Student> students, int[] capacity) {
        int n = students.size();
        int[] assignment = new int[n];
        Arrays.fill(assignment, -1);

        // Sort students based on priority (descending)
        students.sort((a, b) -> b.priority - a.priority);

        for (Student s : students) {
            for (int pref : s.preferences) {
                if (capacity[pref] > 0) {
                    assignment[s.id] = pref;
                    capacity[pref]--;
                    break;
                }
            }
        }
        return assignment;
    }

    // -------- BACKTRACKING ALGORITHM --------
```

```
public static void backtrack(List<Student> students, int[] capacity,
                             int index, int[] current, int score) {

    // Base case: all students assigned
    if (index == students.size()) {
        if (score > bestScore) {
            bestScore = score;
            bestAssignment = current.clone();
        }
        return;
    }

    Student s = students.get(index);

    for (int pref : s.preferences) {
        if (capacity[pref] > 0) {

            // Assign
            current[s.id] = pref;
            capacity[pref]--;

            // Score calculation (higher preference = higher score)
            int newScore = score + (10 - pref);
```

```
            // Recurse
            backtrack(students, capacity, index + 1, current, newScore);

            // Undo (Backtrack)
            capacity[pref]++;
            current[s.id] = -1;
        }
    }
}

// -------- MAIN FUNCTION --------
public static void main(String[] args) {

    int[] capacity = {2, 1}; // Internship capacities

    List<Student> students = new ArrayList<>();

    // Student(id, priority, preferences)
    students.add(new Student(0, 90, Arrays.asList(0, 1)));
    students.add(new Student(1, 80, Arrays.asList(0)));
    students.add(new Student(2, 70, Arrays.asList(1)));

    // ---- GREEDY RESULT ----
    int[] greedyResult = greedy(new ArrayList<>(students), capacity.clone());
```

```
        // ---- BACKTRACKING RESULT ----
        int[] current = new int[students.size()];
        Arrays.fill(current, -1);
        bestAssignment = new int[students.size()];

        backtrack(students, capacity.clone(), 0, current, 0);

        // ---- OUTPUT ----
        System.out.println("Greedy Allocation:");
        print(greedyResult);

        System.out.println("\nOptimal Allocation:");
        print(bestAssignment);
    }

    // -------- PRINT FUNCTION --------
    static void print(int[] arr) {
        for (int i = 0; i < arr.length; i++) {
            System.out.println("Student " + i + " -> Internship " + arr[i]);
        }
    }
}
```

SAMPLE INPUT TABLE

Student	Preference 1	Preference 2	Preference 3
S1	I1	I2	I3
S2	I1	I3	I2
S3	I2	I1	I3

Internship Capacity Table

Internship	**Capacity**
I1	2
I2	1
I3	1

SAMPLE OUTPUT TABLE

STUDENTS	GREEDY ALLOCATION	OPTIMAL ALLOCATION
S1	I1	I1
S2	I1	I3
S3	I2	I2

SCORING SYSTEM TABLE

Preference Level	Score Assigned
1st Choice	3
2nd Choice	2
3rd Choice	1
Not Assigned	0

4.2 Data Structures Used

The implementation of the Smart Allocation Engine relies on fundamental data structures to efficiently manage and process the input data. Arrays are used to track the capacity of each internship, allowing the system to quickly determine whether a particular internship has available slots. Lists are used to store the preference rankings of students, enabling easy access and iteration during the allocation process.

In addition, recursion is used in the Backtracking algorithm to explore all possible assignment combinations. The use of recursion allows the algorithm to systematically traverse the state-space tree, making it easier to implement and understand.

4.3 Algorithm Flow and Execution

The system follows a structured sequence of steps to perform the allocation process. Initially, the input data is collected, including the list of students, their preferences, eligibility information, and the capacity of each internship. This data is then validated to ensure that all constraints are properly defined.

Once the input data is prepared, the Greedy algorithm is executed. Students are first sorted based on priority, and each student is assigned to the highest-ranked internship that has available capacity and satisfies eligibility constraints. This step produces a fast, approximate solution.

Following this, the Backtracking algorithm is executed to find the optimal solution. The algorithm recursively assigns internships to students while ensuring that all constraints are satisfied. If a constraint is violated, the algorithm backtracks and tries a different assignment. This process continues until all possible combinations have been explored.

Finally, the results obtained from both algorithms are compared based on their satisfaction scores and execution times. The system outputs the allocation results along with performance metrics, providing a comprehensive analysis of both approaches.

4.4 Optimization Techniques Used

Several optimization techniques are incorporated into the implementation to improve performance and efficiency. One such technique is the use of sorting, which ensures that higher-priority students are considered first in the Greedy algorithm. This increases the likelihood of achieving a better allocation outcome.

Another important technique is the use of cloning or copying data structures in the Backtracking algorithm. This prevents unintended modifications to the original data during recursive calls, ensuring the correctness of the algorithm.

Additionally, pruning techniques can be applied to reduce the search space in the Backtracking algorithm. By eliminating branches that cannot lead to a better solution, the algorithm can reduce unnecessary computations and improve efficiency.

5. RESULTS AND DISCUSSION

5.1 Experimental Setup

To evaluate the performance of the proposed system, a simulated dataset of students and internships is used. The dataset includes multiple students with varying preference lists and internships with different capacity limits. This setup allows for a comprehensive analysis of how the algorithms perform under different conditions.

5.2 Performance Evaluation

The performance of the Greedy and Backtracking algorithms is evaluated based on several key metrics, including execution time, satisfaction score, and fairness index. The execution time measures how quickly the algorithm produces a solution, while the satisfaction score indicates how well the allocation matches student preferences. The fairness index evaluates how equitably the internships are distributed among students.

The results show that the Greedy algorithm performs significantly faster than the Backtracking algorithm, making it suitable for large datasets and real-time applications. However, the satisfaction score achieved by the Greedy algorithm is slightly lower compared to that of the Backtracking algorithm, which consistently produces the optimal solution.

5.3 Observations and Insights

One of the key observations from the results is that the Greedy algorithm performs well in scenarios where the preference lists are diverse and conflicts are minimal. In such cases, the algorithm is able to assign students to their preferred internships without significant compromises.

However, in scenarios where multiple students have similar preferences, the Greedy algorithm may produce suboptimal results due to its local decision-making nature. In contrast, the Backtracking algorithm is able to handle such conflicts effectively by exploring all possible combinations and selecting the best one.

5.4 Real-World Interpretation

The findings of this study have important implications for real-world applications. The Greedy algorithm can be used in systems that require fast and scalable solutions, such as online internship portals and job matching platforms. On the other hand, the Backtracking algorithm is more suitable for critical decision-making scenarios where achieving the best possible outcome is essential, even if it requires more computation time.

5.5 Key Insight

A major conclusion drawn from this analysis is that no single algorithm is universally optimal for all scenarios. Instead, a hybrid approach that combines the strengths of both Greedy and Backtracking algorithms may provide a more balanced solution, achieving both efficiency and optimality.

CONCLUSION

The Smart Allocation Engine developed in this project successfully demonstrates the application of algorithmic techniques to solve the complex problem of internship allocation. By modeling the problem as a constrained optimization task, the system is able to efficiently assign students to internships while satisfying multiple constraints.

The comparative analysis of the Greedy and Backtracking algorithms highlights the fundamental trade-off between computational efficiency and solution optimality. While the Greedy algorithm offers a fast and scalable solution suitable for large datasets, the Backtracking algorithm guarantees the optimal solution by exhaustively exploring all possibilities.

The integration of a preference-based scoring system and a fairness metric further enhances the effectiveness of the system, providing a more comprehensive evaluation of allocation quality. These features make the system more practical and applicable to real-world scenarios.

Overall, this project provides a strong foundation for the development of intelligent allocation systems that can be used in large-scale applications such as national internship programs and centralized placement platforms. It also demonstrates the importance of algorithm design in solving real-world problems and highlights the potential for further research and development in this area.

FUTURE SCOPE (FULLY EXPANDED)

- The current implementation of the Smart Allocation Engine provides a solid foundation for solving the internship allocation problem. However, there are several opportunities for further enhancement and expansion.
- One potential improvement is the integration of machine learning techniques to predict student preferences and improve allocation accuracy. By analyzing historical data, the system can make more informed decisions and provide better recommendations.
- Another area for future work is the implementation of advanced optimization techniques such as Branch and Bound, which can reduce the computational complexity of the Backtracking algorithm while still maintaining optimality.
- The system can also be extended to support distributed and cloud-based architectures, enabling it to handle large-scale datasets and real-time applications. Additionally, the development of a web-based interface would make the system more accessible and user-friendly, allowing students and administrators to interact with the system more effectively.
- Finally, the inclusion of real-time dashboards and analytics can provide valuable insights into the allocation process, helping decision-makers evaluate system performance and make improvements.

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